Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 8 · SLO 2 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 8 : SLO â€“ 2 | Complexity â€” Time and Space Trade-off

I. Fill in the Blanks

1. Time â€” the amount of time an algorithm takes.
2. Space â€” the amount of memory an algorithm uses.
3. time and space â€” the trade-off balances the two.
4. time â€” more memory (e.g. caching) can reduce time complexity.
5. slowly (longer) â€” using less space may cost more time.
6. memoization (caching) â€” storing precomputed values to save time.
7. operations (elementary steps) â€” how time complexity is measured.
8. input â€” space complexity includes auxiliary space plus input space.
9. space (stack) â€” recursion uses call-stack memory, unlike iteration.
10. constant â€” O(1) space uses a fixed amount of memory.
11. trade-off â€” sorting algorithms show clear time/space trade-offs.
12. iteratively â€” iterative binary search is O(log n) time, O(1) space.
13. computational (algorithmic) â€” both are forms of computational complexity.
14. computations (subproblems) â€” DP stores results to avoid recomputation.
15. balance (trade-off) â€” optimization seeks the best timeâ€“space balance.